

Homework 4

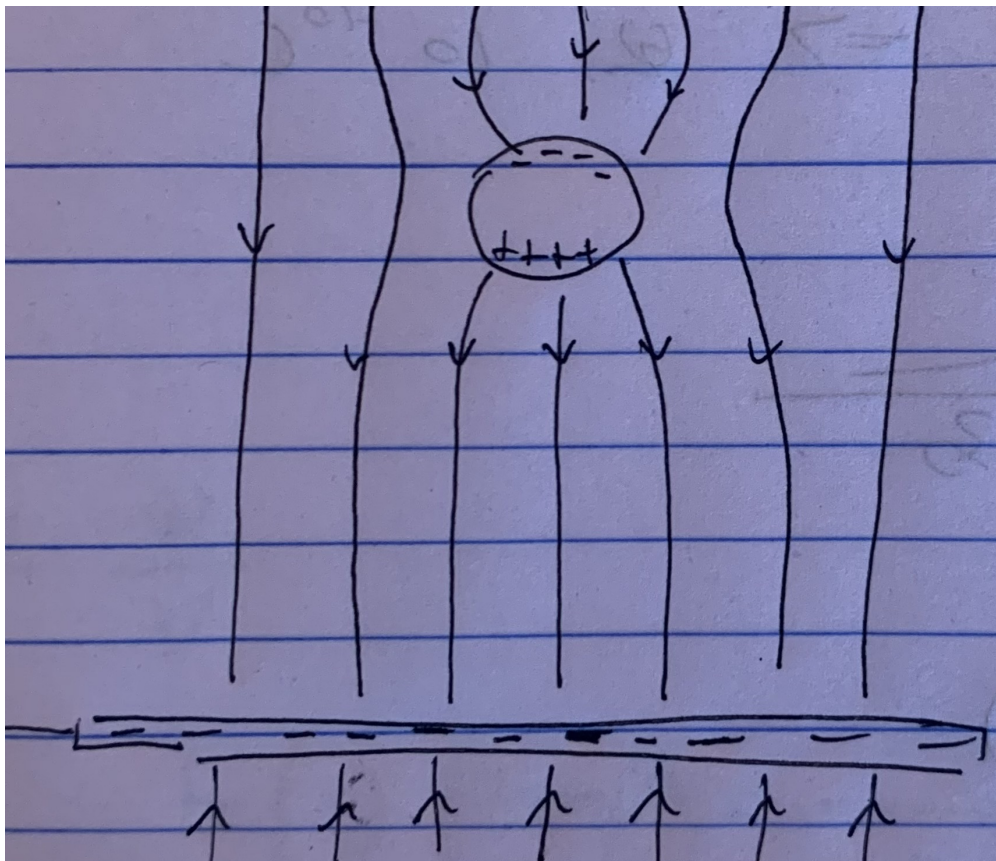
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Physics 16
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May 1, 2026

Question 1. Suppose you have an infinite plane uniformly negatively charged, and above it (some finite distance away) a neutral conducting sphere. Sketch the electric field lines, indicating clearly the direction of the electric field.

Solution. We draw the field lines as such:

□



Question 2. (Purcell ch. 3, ex. 3.59 — *Coaxial capacitor*.) A capacitor consists of two coaxial cylinders of length L , with outer and inner radii a and b . Assume $L \gg a - b$, so that end corrections may be neglected.

(a) Show that the capacitance is

$$C = \frac{2\pi\epsilon_0 L}{\ln(a/b)}.$$

(b) Verify that if the gap between the cylinders, $a - b$, is very small compared with the radius, this result reduces to one that could have been obtained by using the formula for the parallel plate capacitor.

Proof. For part (a), Let the inner cylinder carry charge $Q = \lambda L$. By Gauss's law, taking a cylindrical surface of radius r with $b < r < a$, the enclosed charge per unit length is λ , so

$$E(r) = \frac{\lambda}{2\pi\epsilon_0 r}.$$

The potential difference between the cylinders is

$$V = - \int_a^b \vec{E} \cdot d\vec{l} = \int_b^a E(r) dr = \frac{\lambda}{2\pi\epsilon_0} \int_b^a \frac{dr}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{b}\right) = \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{a}{b}\right).$$

The capacitance is therefore

$$C = \frac{Q}{V} = \boxed{\frac{2\pi\epsilon_0 L}{\ln(a/b)}}.$$

For part (b), when $a - b \ll b$ we write $a/b = 1 + (a - b)/b$. Since $(a - b)/b \ll 1$, the Taylor expansion $\ln(1 + x) \approx x$ gives

$$\ln\left(\frac{a}{b}\right) \approx \frac{a - b}{b}.$$

Substituting,

$$C \approx \frac{2\pi\epsilon_0 L}{\frac{a - b}{b}} = \frac{2\pi b L \epsilon_0}{a - b} = \boxed{\frac{\epsilon_0 A}{d}},$$

where $A = 2\pi b L$ is the area of the inner cylinder and $d = a - b$ is the gap width d . □

Question 3. (Purcell ch. 3, ex. 3.66 — *Adding a capacitor*.) A 100 pF capacitor is charged to 100 volts. After the charging battery is disconnected, the capacitor is connected in parallel with another capacitor. If the final voltage is 30 volts, what is the capacitance of the second capacitor? How much energy was lost, and what happened to it?

Solution. When the battery is disconnected, the charge on $C_1 = 100$ pF is fixed at

$$Q = C_1 V_0 = (100 \text{ pF})(100 \text{ V}) = 10,000 \text{ pC}.$$

When C_2 is connected in parallel, the two capacitors share a common voltage $V_f = 30 \text{ V}$ while total charge is conserved:

$$Q = (C_1 + C_2) V_f.$$

Solving for C_2 ,

$$C_2 = \frac{Q - C_1 V_f}{V_f} = \frac{10,000 - (100)(30)}{30} = \frac{7,000}{30} \approx \boxed{233 \text{ pF}}.$$

The initial and final energies are

$$U_i = \frac{1}{2} C_1 V_0^2 = \frac{1}{2} (100 \times 10^{-12} \text{ F})(100 \text{ V})^2 = 5.0 \times 10^{-7} \text{ J},$$

$$U_f = \frac{1}{2} (C_1 + C_2) V_f^2 = \frac{1}{2} (333 \times 10^{-12} \text{ F})(30 \text{ V})^2 = 1.5 \times 10^{-7} \text{ J}.$$

Therefore,

$$\Delta U = U_f - U_i = \boxed{-3.5 \times 10^{-7} \text{ J}}.$$

This energy is dissipated as heat in the resistance of the connecting wires. □

Question 4. (Purcell ch. 3, prob. 3.20 — *Distribution of charge on a capacitor.*) Consider a parallel-plate capacitor with *different* magnitudes of charge on the two plates. Let the charges be Q_1 and Q_2 (which we normally set equal to Q and $-Q$). Find the four amounts of charge on the inner and outer surfaces of the two plates.

Solution. Label the four surface charge densities as $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ from left to right, corresponding to the outer face of plate 1, the inner face of plate 1, the inner face of plate 2, and the outer face of plate 2. Each surface contributes a field of magnitude $\sigma/2\epsilon_0$ pointing away from it on both sides. By superposition, the total field in each region is the sum of contributions from all four surfaces. The total charge on each plate is fixed as shown by the system of equations

$$\sigma_1 + \sigma_2 = \frac{Q_1}{A}, \quad \sigma_3 + \sigma_4 = \frac{Q_2}{A}.$$

Now, note the field inside plate 1 must be zero since it is a conductor. Taking rightward as positive, the contributions from the four surfaces are $+\sigma_1/2\epsilon_0$, $-\sigma_2/2\epsilon_0$, $-\sigma_3/2\epsilon_0$, $-\sigma_4/2\epsilon_0$, giving

$$\sigma_1 - \sigma_2 - \sigma_3 - \sigma_4 = 0.$$

Similarly, the field inside plate 2 must vanish, giving

$$\sigma_1 + \sigma_2 + \sigma_3 - \sigma_4 = 0.$$

Adding the two zero-field equations gives $2\sigma_1 - 2\sigma_4 = 0$, so $\sigma_1 = \sigma_4$. Subtracting gives $-2\sigma_2 - 2\sigma_3 = 0 \implies \sigma_2 = -\sigma_3$. Substituting back into the charge conservation equations gives

$$\sigma_1 + \sigma_2 = \frac{Q_1}{A}, \quad -\sigma_2 + \sigma_1 = \frac{Q_2}{A},$$

where we used $\sigma_4 = \sigma_1$ and $\sigma_3 = -\sigma_2$. Then, adding gives

$$2\sigma_1 = \frac{Q_1 + Q_2}{A} \implies \sigma_1 = \sigma_4 = \frac{Q_1 + Q_2}{2A},$$

and subtracting gives

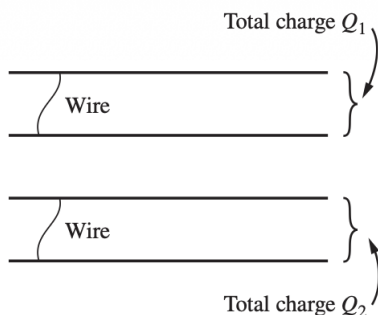
$$2\sigma_2 = \frac{Q_1 - Q_2}{A} \implies \sigma_2 = \frac{Q_1 - Q_2}{2A}, \quad \sigma_3 = -\frac{Q_1 - Q_2}{2A}.$$

Multiplying through by A , the four surface charges are

$Q_{\text{outer},1} = Q_{\text{outer},2} = \frac{Q_1 + Q_2}{2}, \quad Q_{\text{inner},1} = \frac{Q_1 - Q_2}{2}, \quad Q_{\text{inner},2} = -\frac{Q_1 - Q_2}{2}.$

□

Question 5. (Purcell ch. 3, ex. 3.55 — *Two pairs of plates.*) Do the previous problem first. [See Problem 4 above.]

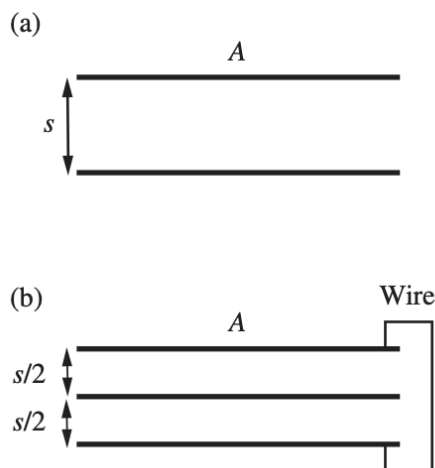


Solution. Since wires are connected between the upper two plates and lower two plates, the potential is zero in-between them. Hence, the problem setup is very similar to Problem 4 because we can translate the four plates in this problem to the four independent surfaces in Problem 4. Hence, the charges on each of the four plates are

$$Q_{1,\text{top}} = \frac{Q_1 + Q_2}{2}, \quad Q_{1,\text{bottom}} = \frac{Q_1 - Q_2}{2}, \quad Q_{2,\text{top}} = -\frac{Q_1 - Q_2}{2}, \quad Q_{2,\text{bottom}} = \frac{Q_1 + Q_2}{2}.$$

□

Question 6. (Purcell ch. 3, ex. 3.53 — *Inserting a plate.*) Consider an existing capacitor of capacitance C . A conducting plate is inserted into the gap. Calculate the new capacitance.



Solution. The original capacitance is $C = \epsilon_0 A/s$. Inserting a thin conducting plate in the middle divides the system into two sub-capacitors, each with gap $s/2$ and area A :

$$C_{\text{top}} = C_{\text{bottom}} = \frac{\epsilon_0 A}{s/2} = 2C.$$

The wire connecting the outer plates forces them to the same potential. The middle plate serves as the opposite terminal for both sub-capacitors. This means the two sub-capacitors share the same pair of terminals (outer plates on one side, middle plate on the other), so they are in parallel, implying

$$C_{\text{net}} = C_{\text{top}} + C_{\text{bottom}} = 2C + 2C = \boxed{4C}.$$

□

Question 7. (Purcell ch. 3, ex. 3.67 — *Energy in coaxial tubes*.) Two coaxial aluminum tubes are 30 cm long. The outer diameter of the inner tube is 3 cm, and the inner diameter of the outer tube is 4 cm. When these are connected to a 45 volt battery, how much energy is stored in the electric field between the tubes?

Solution. The relevant radii are $b = 1.5$ cm and $a = 2.0$ cm (inner and outer, respectively), and the length is $L = 0.30$ m. Using the result of Problem 2, the capacitance is

$$C = \frac{2\pi\epsilon_0 L}{\ln(a/b)} = \frac{2\pi(8.85 \times 10^{-12})(0.30)}{\ln(2.0/1.5)} = \frac{1.67 \times 10^{-11}}{0.2877} \approx 5.80 \times 10^{-11} \text{ F}.$$

With $V = 45$ V, the energy stored is

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(5.80 \times 10^{-11} \text{ F})(45 \text{ V})^2 \approx \boxed{5.9 \times 10^{-8} \text{ J}}.$$

□